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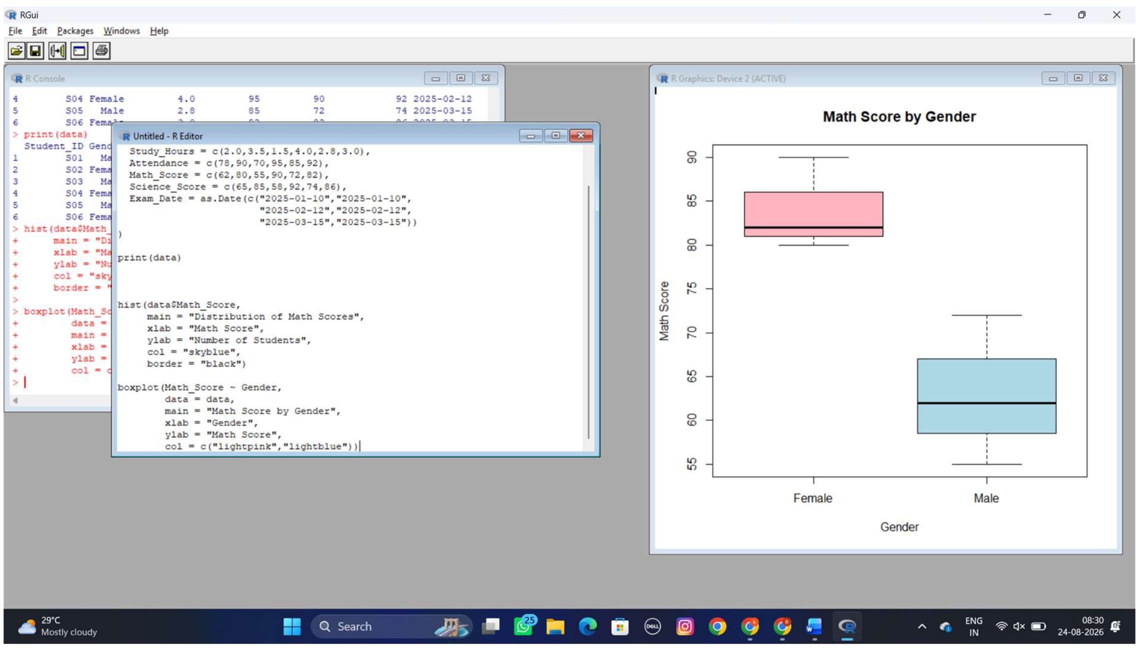
DATA-HANDLING-AND-VISUVALIZATION-LAB_26-30

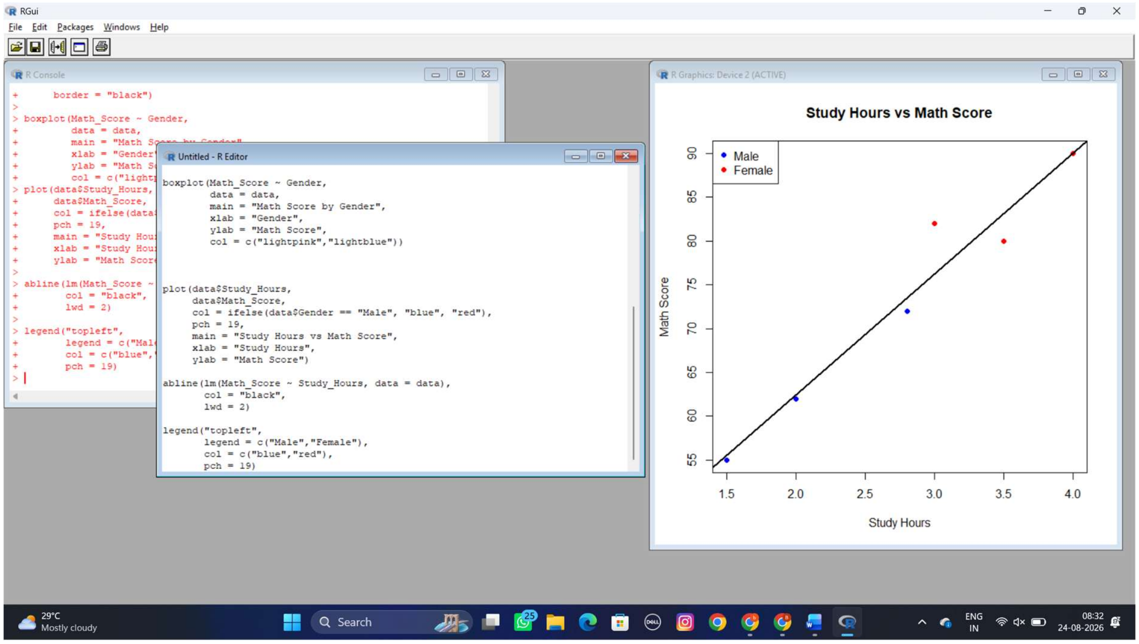
SET 26

Dataset: Student_Mini_Data.csv

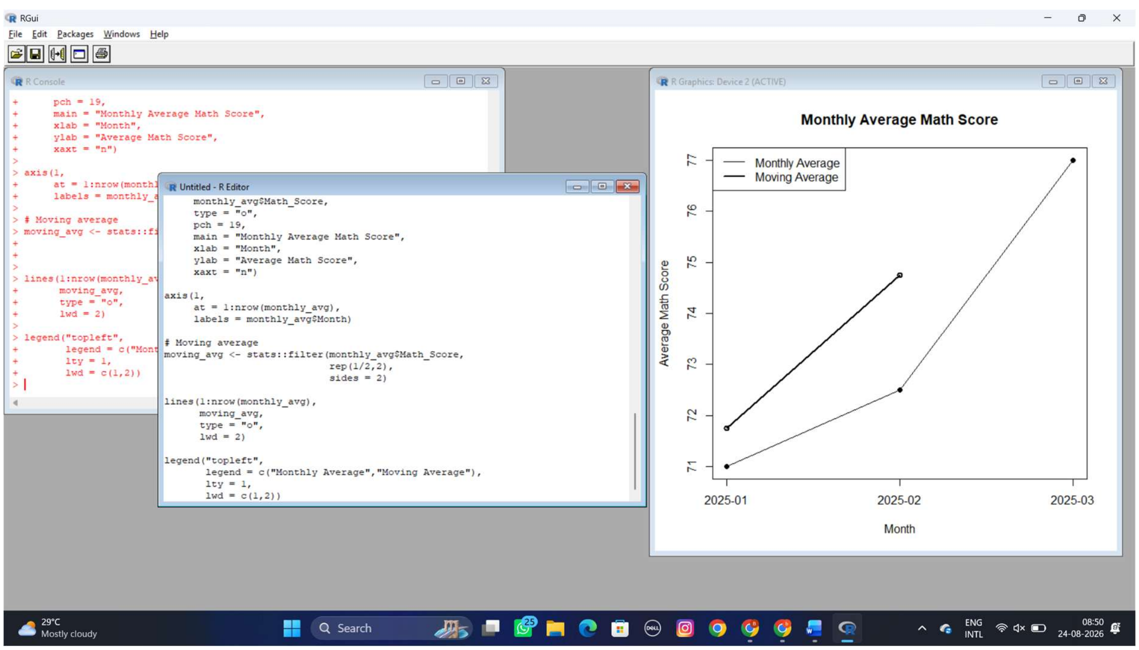
1. **Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.**



2. **Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.**



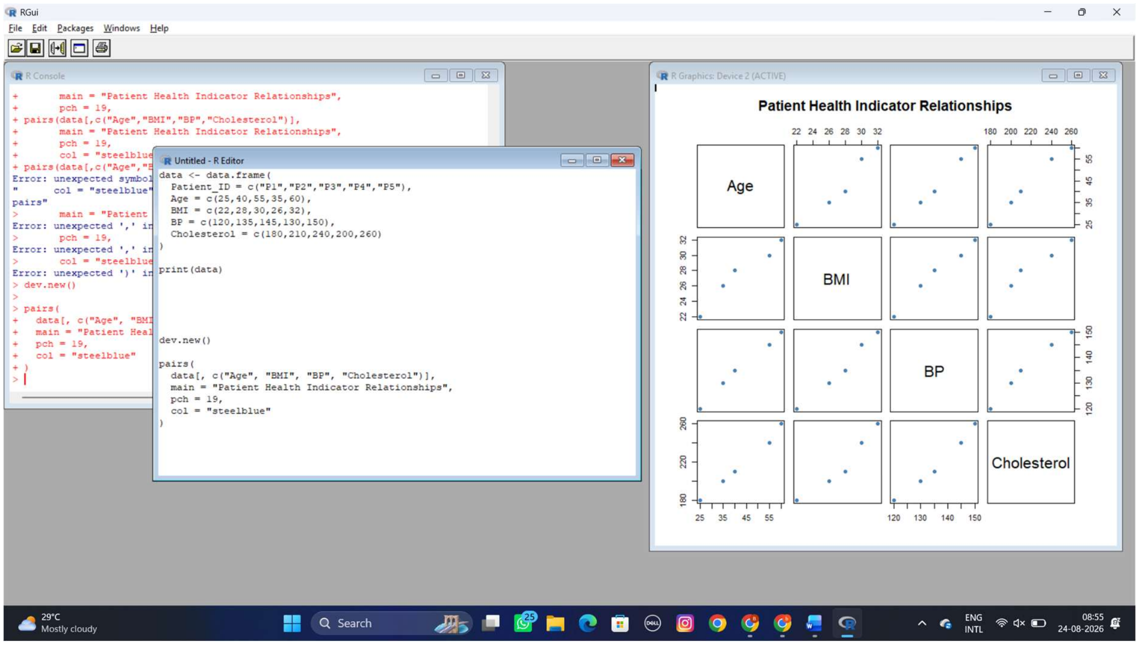
3. **Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.**



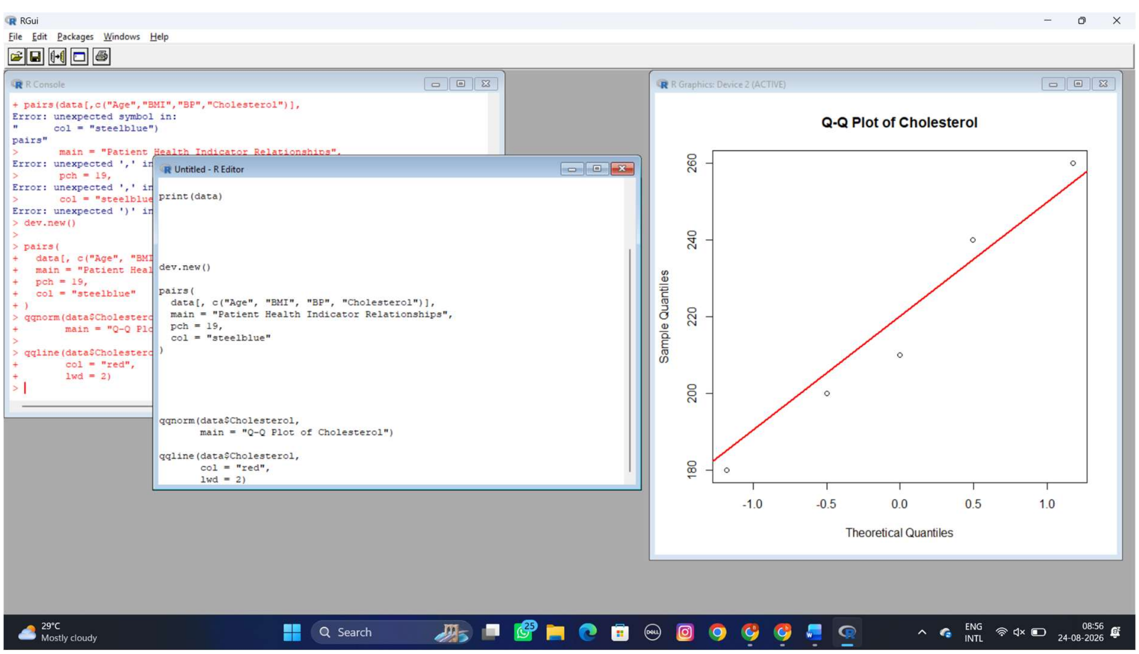
SET 27.

Patient Health Risk Analysis

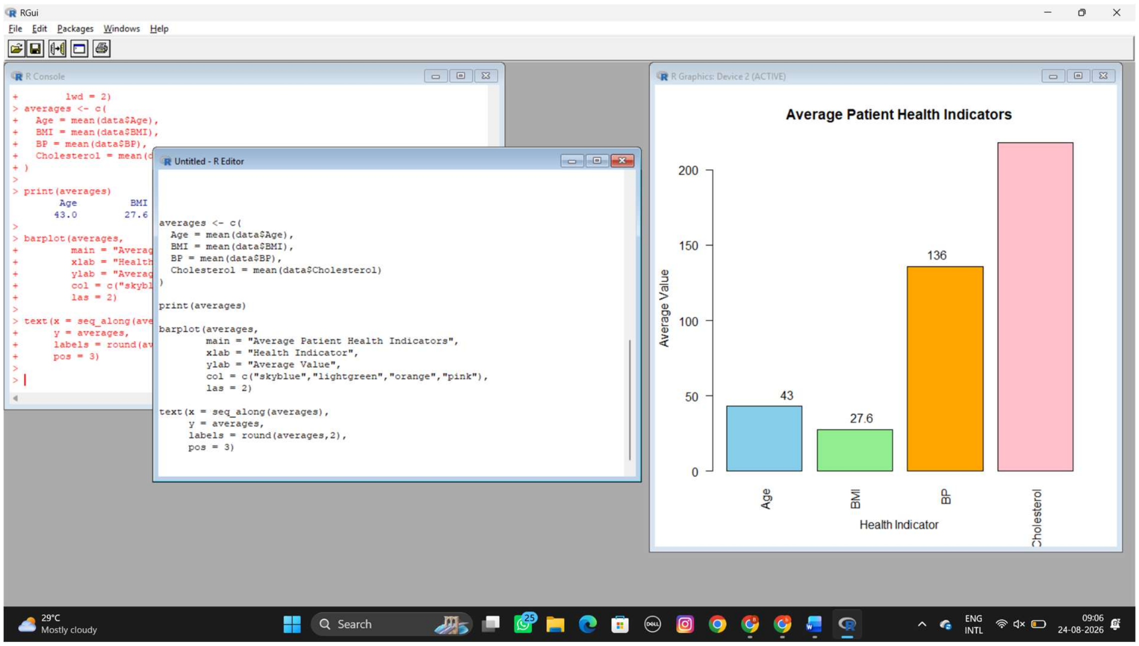
1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol.Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.



2.Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.



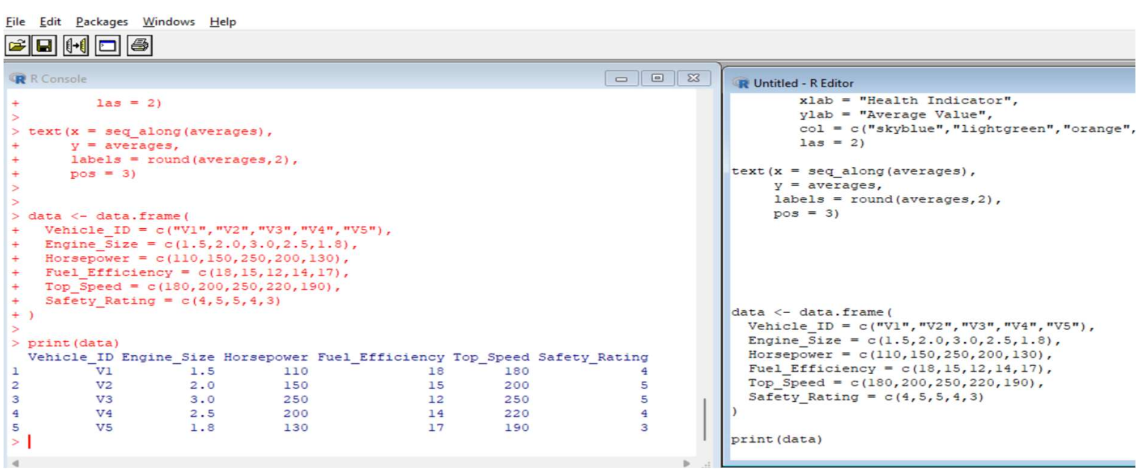
3)Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

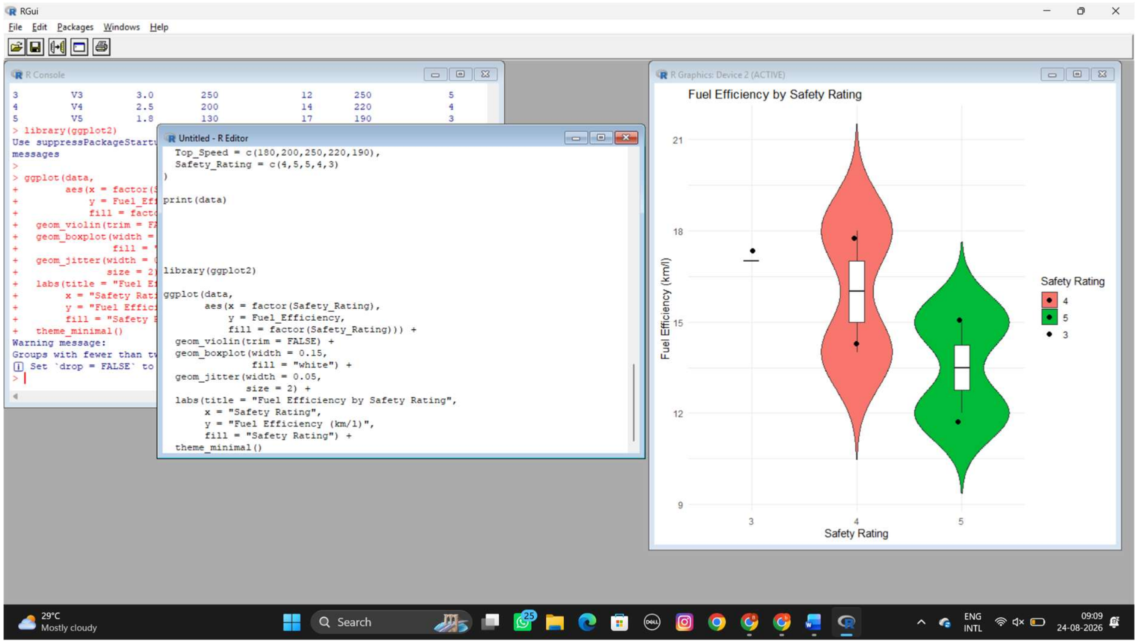


SET 28. Vehicle Performance Analysis

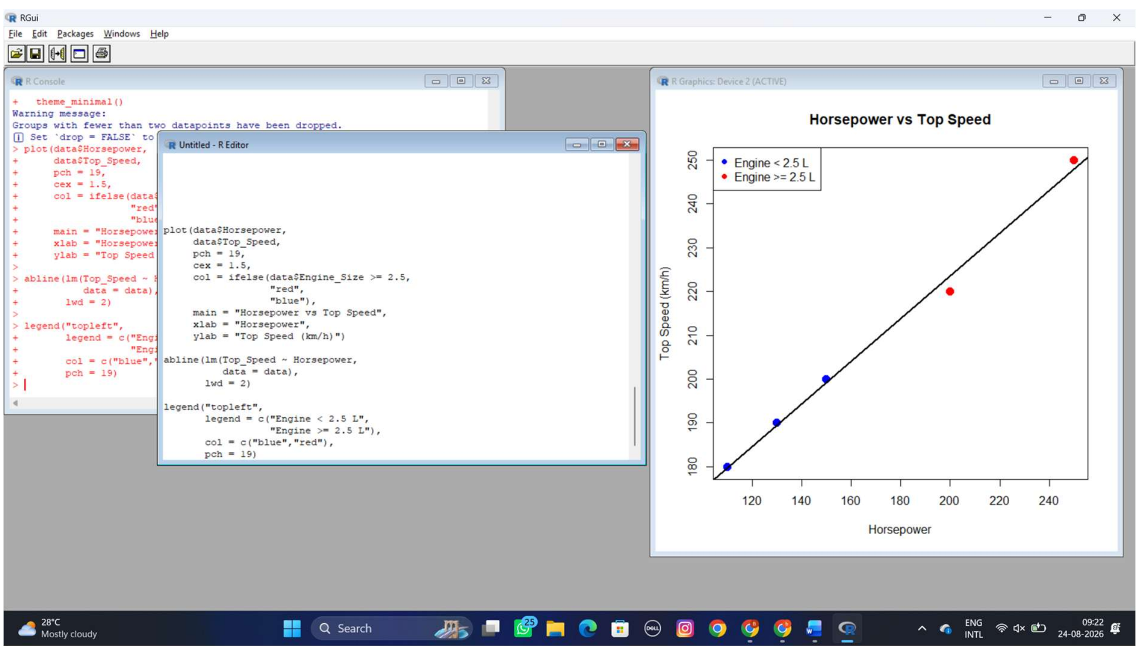
Dataset:

1. **Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.**

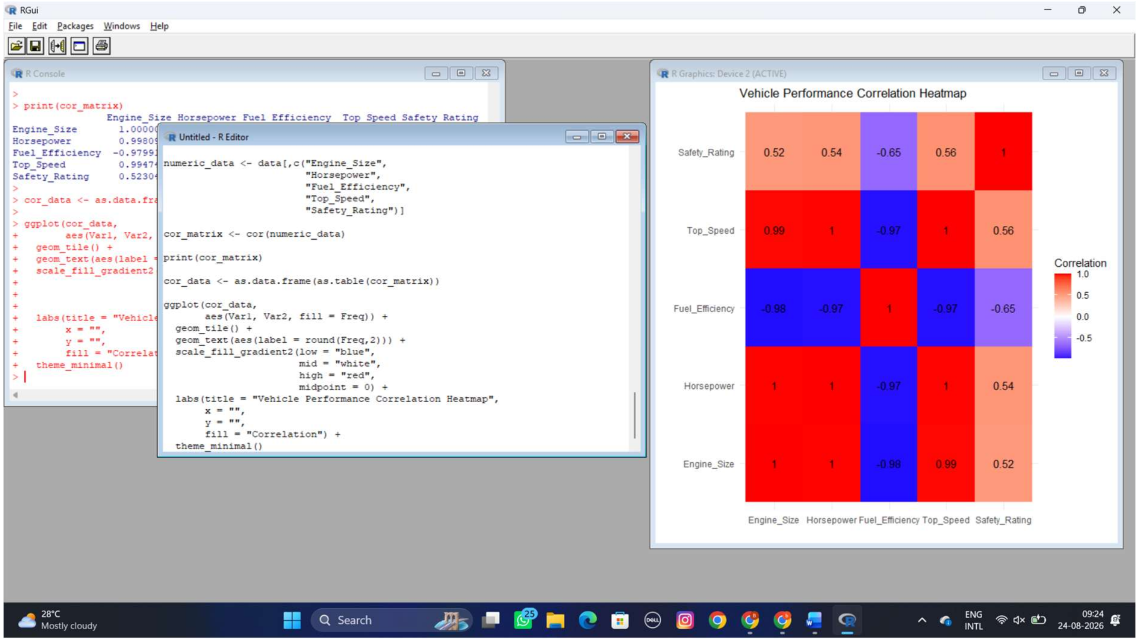




2. **Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.**



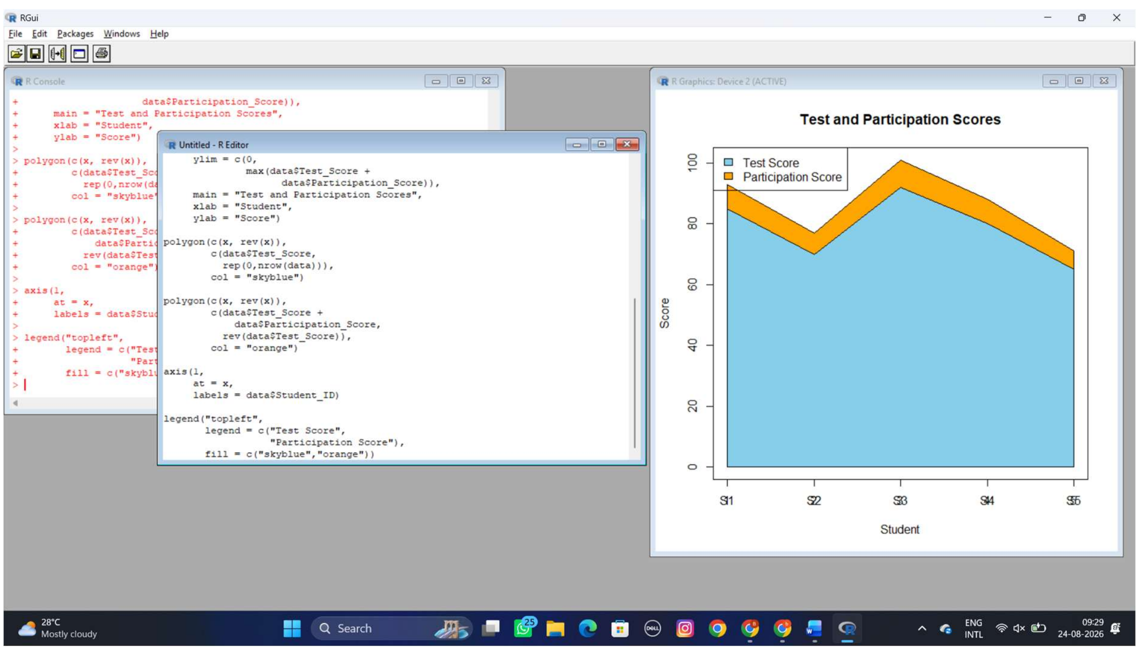
3. **Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.**



SET 29 .Student Academic Performance

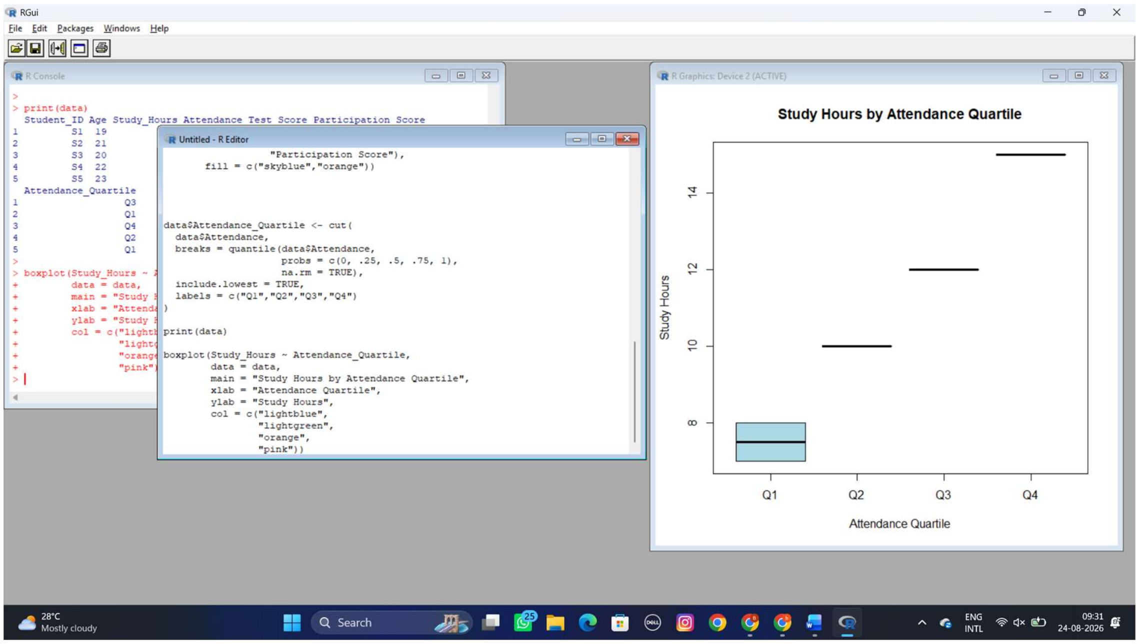
Dataset:

1. **Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.**

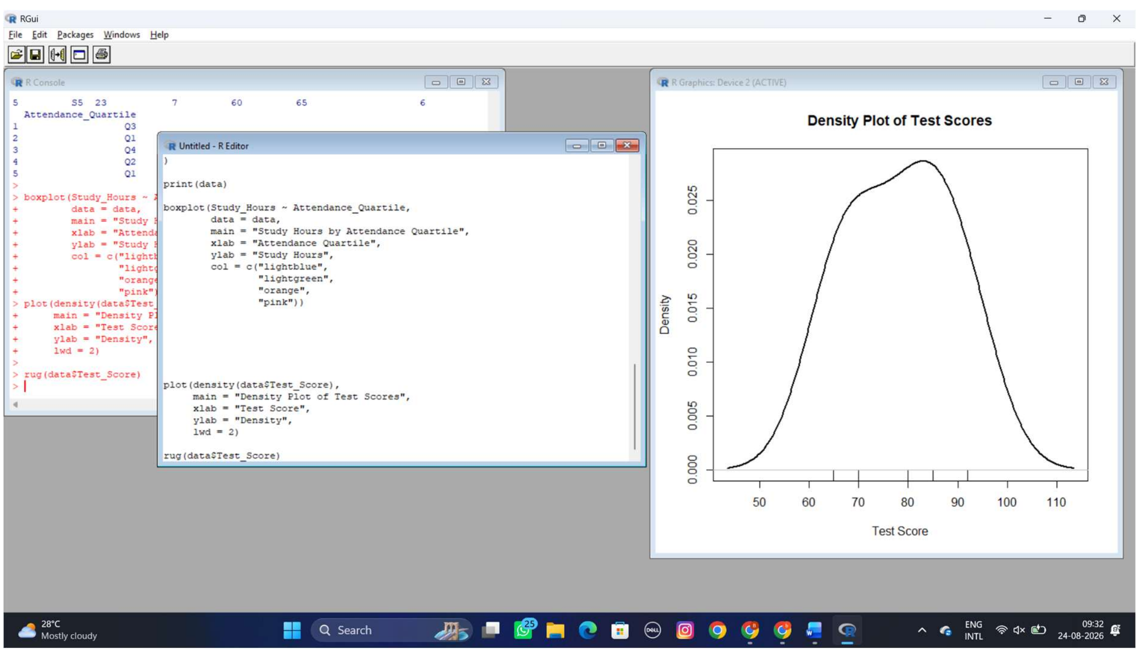


2. **Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and**

interquartile range of study hours per attendance quartile and any outliers in study hours.



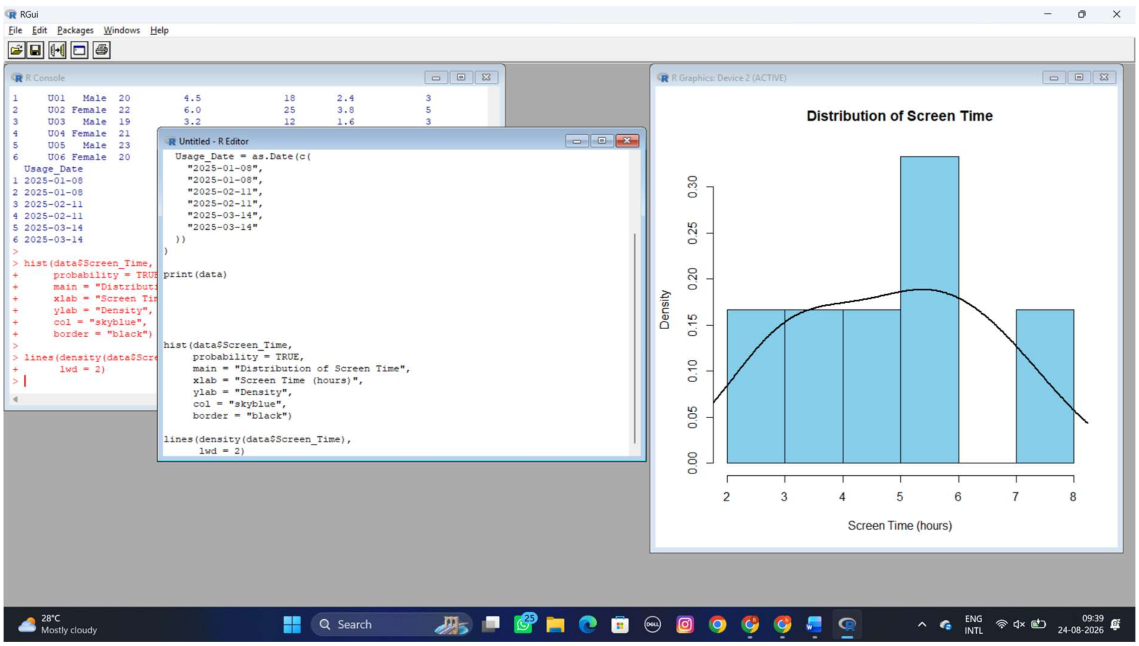
3. **Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.**



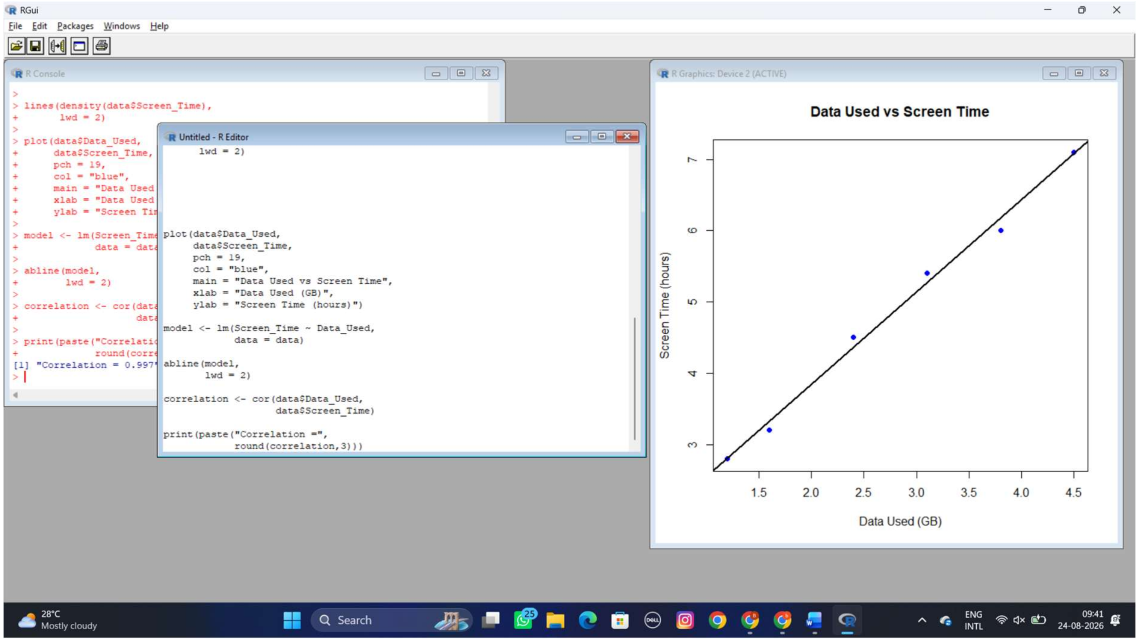
SET 30

Dataset: Mobile_App_Usage.csv

1. **Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.**



2. **Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.**



3. **Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.**